

A Study into Cognitive Frameworks, Decision-Making Mechanisms, and Neural Specialisation

A comprehensive briefing synthesising research on dual-process cognition, heuristic biases, functional brain specialisation, and the enduring debate over conscious agency with implications for clinical and legal practice.

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EXECUTIVE BRIEFING

COGNITIVE NEUROSCIENCE



Critical Takeaways: 1 of 2

1

Dual-Process Theory

Kahneman's framework posits two cognitive systems: System 1 is fast, automatic, and unconscious, handling most daily activity but prone to bias; System 2 is slow, deliberate, and effortful, engaging for complex reasoning. System 1 is the default, so its outputs are usually accepted unless System 2 actively overrides them.

2

Heuristics and Anchoring Effect

Tversky and Kahneman showed that an initial value can systematically skew later estimates, even when the anchor is arbitrary. Related heuristics compound this: availability inflates vivid events, representativeness produces the conjunction fallacy, framing changes choices, and sunk costs sustain failing endeavours.

3

Neural Modularity

Discrete cortical regions serve specific functions: the FFA recognises faces, V5/MT detects motion, V4 processes colour, the PPA encodes spatial environments, and the prefrontal cortex governs executive control. Split-brain research confirmed hemispheric specialisation. Left for language and analysis, Right for spatial and holistic processing.

4

Freud and the Unconscious

Freud's structural model divides the psyche into Id, Ego, and Superego. The unconscious shapes behaviour through repression, displacement, and projection. Libet's EEG experiments showed readiness potential precedes conscious intention by about 350ms, suggesting voluntary actions begin unconsciously.

Critical Takeaways: 2 of 2

1

Jung and the Collective Unconscious

Jung proposed a collective unconscious shared across humanity, populated by archetypes such as the Persona, Shadow, Anima/Animus, and Self. Individuation integrates these elements into awareness and represents psychological maturity. He also introduced introversion/extraversion and the complex.

2

Freud vs. Jung: The Great Divide

Their collaboration ended in rupture over deep theoretical disagreements: Freud treated libido as sexual and the Oedipus complex as universal; Jung reframed libido as general psychic energy and rejected sexual reductionism. Freud's model was mechanistic, while Jung's was teleological and meaning-centred.

3

Replicability Crisis

Social priming studies, including Bargh's elderly priming work, failed to replicate in multiple independent attempts. Analysis found 11 of 12 key studies had an R-Index below 50%, pointing to inflation from small samples and publication bias. The crisis drove reforms such as pre-registration, larger samples, open data, and adversarial collaboration.

4

Clinical and Legal Implications

The M'Naghten Rules exempt defendants where a disease of the mind prevents knowledge of the act or its wrongness, mapping directly onto System 2 failure. Diminished responsibility can reduce murder to manslaughter, and syndromes like Anton Syndrome, Riddoch Syndrome, and prosopagnosia show how modular perception affects capacity, intent, and culpability.



CHAPTER I

The Dual-Process Model: System 1 and System 2

Popularised by Daniel Kahneman, these are not physical brain structures but **functional descriptions** of how the mind manages information. One fast and effortless, the other deliberate and demanding.

Feature	System 1	System 2
Speed	Fast	Slow
Effort	Low / None	High
Awareness	Unconscious	Conscious
Control	Automatic	Intentional
Role	Intuition & Impulse	Rational Reasoning & Self-Control

System 1: Fast & Intuitive

Rooted in associative reasoning and prior experience. Handles habitual tasks, emotional responses, and pattern recognition, but generates systematic biases when over-relied upon.

System 2: Slow & Analytical

Activated when System 1 encounters a situation that violates its internal model . As illustrated by the "Invisible Gorilla" experiment, where focused attention causes observers to miss an obvious stimulus. System 2 acts as monitor and editor.

System 1 vs System 2: Real-World Examples

Kahneman illustrates the two systems not through abstract theory but through vivid, relatable examples. The contrast between them reveals how effortlessly System 1 operates, and how rarely System 2 is called upon to intervene.

⚡ System 1 in Action

2 + 2 = ?

The answer arrives instantly, without effort. You did not choose to calculate it, System 1 retrieved it automatically from memory.

The Angry Face

You detect hostility in someone's expression before you consciously register their features. System 1 reads emotional cues faster than System 2 can process them.

Driving on a Familiar Road

An experienced driver navigates a known route while holding a conversation. The driving is fully automated, System 1 handles it without conscious direction.

The Bat and Ball Problem

A bat and ball cost \$1.10. The bat costs \$1 more than the ball. How much is the ball? System 1 instantly answers "10 cents", which is wrong. The correct answer is 5 cents. System 1 substituted an easier question.

Completing 'Bread and...'

Your mind automatically supplies 'butter'. Associative memory, a hallmark of System 1, fills in the blank without any deliberate thought.

🧠 System 2 Stepping In

17 × 24 = ?

Unlike simple arithmetic, this requires deliberate, step-by-step calculation. You feel the mental effort, your attention narrows, your pace slows. This is System 2 at work.

Parallel Parking in a Tight Space

A novice driver must consciously think through every movement. System 2 takes over when a task exceeds the automation threshold of System 1.

Filling Out a Tax Return

Complex, rule-governed tasks with no automatic solution demand sustained System 2 engagement. Fatigue and distraction degrade performance, System 2 is resource-limited.

Monitoring Your Behaviour at a Party

Choosing words carefully, suppressing inappropriate reactions, maintaining social composure, all require active System 2 override of System 1 impulses.

Searching for a Woman in a Crowd

Kahneman uses this as a classic example of directed attention, you must deliberately scan, filter, and compare, rather than relying on automatic recognition.

📌 **The Lazy Controller:** Kahneman's central insight is that System 2 is "lazy", it endorses System 1's outputs with minimal scrutiny unless explicitly prompted. This is why cognitive biases persist even in intelligent, educated people. The solution is not to distrust intuition entirely, but to learn when to pause and engage System 2, particularly in high-stakes, unfamiliar, or emotionally charged situations.





CHAPTER II

Heuristics and the Anchoring Effect

Anchoring is among the most measurable and influential cognitive heuristics. Tversky and Kahneman demonstrated that adjustment from an initial value is consistently **insufficient**, leaving final estimates biased toward the anchor even when that anchor is arbitrary.



Product Positioning

Starbucks lists premium items such as Evian water at high prices to make core beverages appear more reasonable — the anchor reframes the consumer's reference frame.



Purchasing Limits

When a "limit of 12 per person" was posted on Campbell Soup, shoppers bought **twice as many** cans as those with no limit. The numeral "12" served as an anchor for quantity judgment.



Menu Engineering

Restaurants list items in descending price order. The first, most expensive item anchors perception, making subsequent options appear as prudent savings rather than expenditure.

📌 **Measurability in professional contexts:** Real estate agents assessing house values were influenced by listed prices even when they claimed to disregard them. Anchor index: **41%** for professionals, **48%** for students. In charitable donation studies, requesting \$400 raised average donations to \$143, versus \$20 when \$5 was requested.



CHAPTER II (CONTINUED)

Other Key Heuristics & Cognitive Biases

Beyond anchoring, Kahneman and Tversky identified a suite of systematic biases that distort judgment across professional, legal, and everyday contexts. Each arises from System 1's tendency to substitute hard questions with easier ones, often without our awareness.



Confirmation Bias

We instinctively seek, favour, and recall information that confirms pre-existing beliefs. System 1 filters incoming data to protect our existing worldview. System 2 rarely intervenes to challenge it. Kahneman links this to 'what you see is all there is' (WYSIATI).



Conjunction Fallacy

People judge a specific scenario as more probable than a general one when it fits a compelling narrative. In Kahneman's famous 'Linda Problem', most participants rated 'Linda is a bank teller and feminist' as more likely than 'Linda is a bank teller' violating basic probability rules.



Sunk Cost Fallacy

Past irrecoverable investments, time, money, effort that irrationally influence future decisions. System 1 frames abandonment as loss, compelling continued investment in failing endeavours. Rational decision-making should consider only future costs and benefits.



Framing Effects

Identical information yields different decisions depending on how it is presented. '90% survival rate' and '10% mortality rate' are logically equivalent, yet patients consistently prefer the former. Kahneman demonstrated this across medical, financial, and legal contexts.

 **Availability Heuristic (bonus):** We judge the likelihood of events by how easily examples come to mind. Plane crashes feel more dangerous than car accidents because they are more vivid and memorable even though statistically the reverse is true. Tversky & Kahneman (1973) showed this leads to systematic overestimation of dramatic but rare events.

Scientific Integrity and the Priming Crisis

Social priming is the idea that subtle environmental cues can unconsciously alter behaviour. For example, reading words associated with old age, such as "wrinkle", "grey", or "bingo", was claimed to make participants walk more slowly down a corridor. This finding became a high-profile casualty of the replicability crisis when independent researchers could not reproduce it.



The R-Index Findings

A meta-scientific analysis of Chapter 4 of *Thinking, Fast and Slow* found that **11 of 12 cited priming studies** had an R-Index below 50, indicating very low replicability and probable inflation of effect sizes through selective reporting. For instance, the "Florida Effect" study by Bargh et al. (1996), which claimed elderly-related words slowed walking speed, failed to replicate in a double-blind replication by Doyen et al. (2012). The effect disappeared entirely when experimenters were unaware of the condition.



Kahneman's "Train Wreck" Warning

In 2012, Kahneman sent an open email to leading priming researchers warning of a looming train wreck for the field. He later publicly admitted he had placed too much faith in underpowered studies. A notable example: the "money priming" studies by Vohs et al., which claimed that exposure to images of money made people more selfish and self-sufficient, also failed large-scale replication attempts. Kahneman's candour represents a rare act of scholarly self-correction.



The Scientific Lesson

The failure of social priming studies highlights the dangers of selection for significance, which means reporting only successful trials while filing away null results. This is known as the "file drawer problem". For example, if 20 researchers test the same hypothesis and only the one who gets a positive result publishes, the literature becomes misleadingly optimistic. Robust inference requires large sample sizes, pre-registration of hypotheses, and independent replication before publication.



Flags & Voting Behaviour

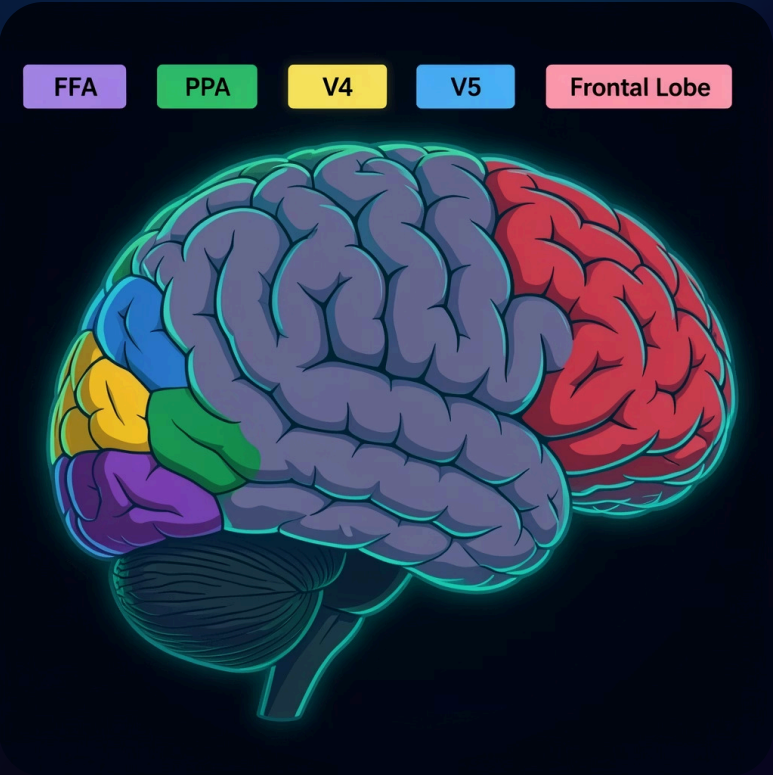
Research by Carter and McVee (2011) claimed that brief exposure to an American flag image shifted participants' political attitudes toward Republican positions, a striking example of symbolic priming. The study suggested that even a fleeting glimpse of a national symbol could unconsciously bias political judgment. However, like many social priming findings, the effect proved fragile: subsequent large-scale replications failed to reproduce the shift, and the study was cited in the Open Science Collaboration's 2015 reproducibility report as one of many high-profile failures. It illustrates how politically and culturally charged priming claims attracted outsized media attention, and outsized scrutiny.

Functional Specialisation of the Brain

Hemispheric Specialisation

Established through Roger Sperry's landmark **split-brain experiments** on callosotomy patients, revealing independent processing in each hemisphere:

Left Hemisphere Dominant in language, verbal cognition, and sequential logic. Controls the right visual field and right hand.	Right Hemisphere Dominant in spatial reasoning, visual imagery, and creative processing. Controls the left visual field and left hand.
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Each of these regions exemplifies the principle of functional specialisation: the brain does not process information as a unified whole, but through a network of dedicated modules.

The FFA and PPA handle identity and space respectively; V4 and V5 decode the visual qualities of colour and motion; and the frontal lobes integrate these inputs into deliberate, goal-directed behaviour.

Damage to any single region produces a highly specific deficit, leaving all other functions intact, which is precisely what makes lesion studies so powerful as evidence for modularity.

Fusiform Face Area (FFA)

Located in the inferior temporal cortex, the FFA responds selectively to faces over all other visual stimuli. Damage produces prosopagnosia, the inability to recognise familiar faces, even those of close family members, while other object recognition remains intact.

Parahippocampal Place Area (PPA)

The PPA encodes spatial layouts, environmental scenes, and navigational contexts. It responds strongly to images of places and buildings, independent of the objects within them. Damage impairs scene recognition and spatial navigation without affecting face or object perception.

V4 — Colour Perception

Area V4 in the occipital lobe is specialised for processing colour and intermediate visual features. Damage results in cerebral achromatopsia, a complete loss of colour vision in which the world appears in shades of grey, while form and motion perception remain unaffected.

V5 — Motion Detection

Area V5, also known as MT, is dedicated to detecting and processing visual motion. Lesions cause cerebral akinetopsia, a rare condition in which the patient cannot perceive movement. Objects appear to jump between static positions rather than moving fluidly, making everyday tasks such as pouring liquid extremely difficult.

Frontal Lobes

The prefrontal cortex governs executive functions including planning, impulse control, working memory, and moral reasoning. Damage, as seen in the famous case of Phineas Gage, can dramatically alter personality and decision-making while leaving basic cognition and motor function largely intact.



CHAPTER IV

The Unconscious Mind and the Free Will Debate

The interplay between conscious intention and unconscious drive remains a central and unresolved conflict in both psychology and neuroscience spanning Freudian theory to contemporary electrophysiology.

The Freudian Tripartite Model

The Id

Primitive and unconscious, governed by the *pleasure principle*. Driven by Eros (life instinct) and Thanatos (death/aggression instinct).

The Ego

Rational mediator operating on the *reality principle*. Manages Id demands within the constraints of social reality via defence mechanisms (denial, projection, sublimation).

The Superego

The internalised moral code and ideal self, functioning as conscience and the seat of guilt.

The Libet Experiment

Benjamin Libet recorded a **"readiness potential"**, measurable neural activity, occurring approximately **350 ms before** a participant consciously decided to move a wrist.

This temporal gap led many to conclude that the brain initiates voluntary action unconsciously, challenging free will as traditionally conceived.

❏ **Counter-argument:** Kevin Mitchell's "emergent agency" framework proposes that the whole organism and not isolated neurons functions as a coherent causal agent. Critics also note Libet's tasks were trivial; deliberative choices show markedly different patterns.

Carl Jung & the Architecture of the Unconscious

Where Freud mapped the unconscious as a reservoir of repressed drives, Carl Jung expanded it into a transpersonal domain shared by all humanity. Jung's Analytical Psychology introduced structural concepts of the psyche that remain influential in clinical, cultural, and organisational psychology.

The Structure of the Psyche

The Persona

The social mask worn to meet external expectations. It is not the true self. Over-identification with the Persona leads to a loss of authentic identity and inner disconnection.

The Shadow

The unconscious repository of traits, impulses, and desires the ego refuses to acknowledge. Jung argued that integrating, not suppressing the Shadow is essential to psychological wholeness.

Anima / Animus

The unconscious feminine aspect in men (Anima) and masculine aspect in women (Animus). These contrasexual archetypes mediate access to the deeper unconscious and influence relational patterns.

The Self

The central archetype representing psychic wholeness and integration. The Self is the goal of individuation. It is the lifelong process of reconciling conscious and unconscious elements into a unified personality.

The Collective Unconscious & Individuation

Unlike Freud's personal unconscious built from individual repressed experience, Jung proposed a deeper layer: the **collective unconscious**, a universal psychic substrate shared across all humans and cultures. It contains **archetypes**: inherited patterns of thought, image, and behaviour that manifest in myths, dreams, and symbols across civilisations.

The process of **individuation** — Jung's equivalent of psychological maturity involves confronting the Shadow, integrating the Anima/Animus, and moving beyond the Persona toward authentic selfhood. Jung saw this not as a cure but as a lifelong developmental journey.

📌 **Jung vs. Freud:** While Freud viewed the unconscious as primarily a source of pathology (repressed trauma and drives), Jung saw it as a creative and compensatory force. Dreams, for Jung, were not disguised wish or fulfilments but meaningful communications from the unconscious that is guiding the individual toward wholeness. This distinction marks a fundamental divergence in depth psychology.



Freud vs. Jung: A Theoretical Divergence

Sigmund Freud and Carl Jung were close collaborators for several years before their theoretical disagreements led to a permanent split in 1913. While both pioneered the study of the inner mind, their views diverged on the fundamental nature of human motivation, the structure of the unconscious, and the goals of psychological development.

The Five Key Points of Divergence

The Nature of the Unconscious Freud saw the unconscious as a repository for repressed memories, trauma, and unacceptable desires. Jung agreed on the personal unconscious but proposed a deeper universal layer, the collective unconscious, containing archetypes shared across all humanity.	Human Motivation & Libido Freud viewed libido as primarily sexual energy driving all behaviour, including creativity (sublimation). Jung reframed libido as a generalised life force encompassing the need for meaning, purpose, and spiritual growth, not reducible to sexuality.
Personality Development Freud was deterministic, focusing on early childhood psychosexual stages (e.g., Oedipus complex). Jung emphasised lifelong development through individuation, integrating conscious and unconscious elements to achieve wholeness at any life stage.	Dream Analysis Freud believed dreams conceal meaning as disguised wish-fulfilments rooted in past trauma. Jung believed dreams reveal meaning, acting as psychological compensation and prospective guidance from the unconscious toward future growth.

The Therapeutic Relationship

Freudian psychoanalysis positions the therapist as a neutral analyst uncovering the past through transference. Jungian therapy is more collaborative, therapist and patient jointly explore present and future symbolic meanings.

Similarities & Differences at a Glance

Feature	Freud	Jung
Unconscious Mind	Personal: repressed memories & drives	Personal + Collective, archetypes & universal symbols
Primary Motivation	Sexual/aggressive libido	Generalised life force; meaning & spirituality
Personality Development	Determined by early childhood	Lifelong individuation process
Dream Interpretation	Conceals repressed wishes (past)	Reveals compensatory guidance (future)
Therapeutic Stance	Neutral analyst; transference focus	Collaborative; symbolic & prospective
Shared Belief	Both acknowledge the unconscious shapes behaviour	Both use talk therapy as primary treatment

 **The Parting of Ways:** The split in 1913 was both intellectual and personal. Jung's 'Psychology of the Unconscious' (1912), which reframed libido and challenged Freud's sexual theory, was the breaking point. Freud reportedly said Jung's departure was 'a loss for science.' Jung later described Freud as a great man 'in the grip of his daimon.' Their divergence gave birth to two distinct schools of depth psychology that continue to shape clinical practice today.



CHAPTER VI

Clinical and Legal Implications

The neuroscience and psychology of cognition carry direct, substantive consequences for legal responsibility and clinical diagnosis.

The M'Naghten Rules (1843)

The foundational legal standard for criminal insanity requires that a defendant, due to a **"disease of the mind,"** either:

1. Did not know the nature and quality of the act being committed, **or**
2. Did not know that the act was wrong.

This maps directly onto a catastrophic failure of System 2 control and Ego-mediated reality-testing, as seen in florid psychotic states where the capacity for deliberate moral reasoning is fundamentally compromised.

Occipital Lobe — Clinical Syndromes

Despite being the smallest cortical lobe, occipital damage produces striking dissociations between awareness and visual function:

Anton Syndrome

Cortical blindness with anosognosia: the patient confabulates vivid visual experiences, unaware of their blindness.

Riddoch Syndrome

Paradoxical ability to perceive moving objects within a blind visual field while remaining unable to detect stationary stimuli.

Occipital Lobe Epilepsy

Characterised by visual hallucinations and rapid eye blinking, typically triggered by flickering or photic stimuli.



CHAPTER VI (CONTINUED)

Diminished Responsibility in the Courtroom: Landmark Cases

The legal doctrine of diminished responsibility, codified in s.2 of the Homicide Act 1957 (UK), allows a murder charge to be reduced to manslaughter where a defendant's 'abnormality of mental functioning' substantially impaired their ability to understand their conduct, form rational judgment, or exercise self-control. The following cases illustrate how cognitive and psychiatric evidence has shaped judicial outcomes across jurisdictions.



R v Peter Sutcliffe (UK, 1981)

The 'Yorkshire Ripper' case became a defining moment for diminished responsibility in English law. All four psychiatrists agreed Sutcliffe suffered from paranoid schizophrenia, hearing divine voices commanding the killings. The prosecution rejected a manslaughter plea; the jury convicted of murder. The case exposed the tension between psychiatric consensus and jury discretion, and Sutcliffe was later transferred to Broadmoor psychiatric hospital, effectively the outcome the defence had sought.



Andrea Yates (USA, 2001 / 2006)

Yates drowned her five children during a severe psychotic episode linked to postpartum schizophrenia. Her 2002 murder conviction was overturned on appeal after false expert testimony was identified. At retrial in 2006, a Texas jury found her not guilty by reason of insanity; she was committed to a high-security psychiatric hospital. The case placed the M'Naghten Rules under intense public scrutiny and reignited debate on the irresistible impulse test.



United States v. John Hinckley Jr. (USA, 1982)

Hinckley shot President Reagan and three others in an attempt to impress actress Jodie Foster. Defence psychiatrists testified he suffered from process schizophrenia and major depressive disorder. The jury returned a not guilty by reason of insanity verdict, triggering a national backlash and the Insanity Defense Reform Act 1984, which raised the burden of proof for insanity defences across US federal courts.



R v Joyce & Kay (UK, 2017)

Both defendants suffered from diagnosed paranoid schizophrenia and killed whilst in acute psychotic episodes triggered by alcohol and drug use. The Court of Appeal confirmed that where an underlying recognised mental condition substantially impairs responsibility, voluntary intoxication does not automatically negate the diminished responsibility defence, a significant clarification of the Homicide Act 1957 as amended by the Coroners and Justice Act 2009.



Daryati v Public Prosecutor (Singapore, 2022)

An Indonesian domestic helper convicted of murder successfully appealed to the Singapore Court of Appeal on grounds of diminished responsibility. The court found she was suffering from persistent depressive disorder with intermittent major depressive episodes at the time of the killing, satisfying all three limbs of the defence under the Penal Code. The conviction was reduced from murder to culpable homicide not amounting to murder.

 The Neuroscience Frontier: Courts are increasingly confronted with neuroimaging evidence (fMRI, PET scans) as part of diminished capacity arguments. In *Roper v. Simmons* (US, 2005), the Supreme Court cited neuroscientific evidence of adolescent brain immaturity to abolish the death penalty for juvenile offenders, a landmark moment where brain science directly shaped constitutional law. As neuroimaging becomes more sophisticated, the boundary between legal responsibility and neurological determinism will only grow more contested.

Closing Reflections

Across the above studies, a single idea has steadily come into focus: the mind is neither a perfectly unified rational agent nor a blank slate, but a layered, highly specialised system. Dual-process theory showed how fast intuitions and slower reflective thought work in tandem; Kahneman and Tversky's heuristics and biases revealed how anchoring, confirmation bias, the conjunction fallacy, sunk cost reasoning, framing effects, and availability can distort judgment; and neural specialisation reminded us that cognition is distributed across distinct brain regions, each bringing its own strengths and vulnerabilities. Taken together, these ideas portray human thought as structured, adaptive, and still prone to predictable error.

The depth psychology tradition deepened that picture by moving below the threshold of awareness. Freud mapped the unconscious as a force shaping desire, conflict, and self-understanding, while Jung expanded the terrain through the collective unconscious and archetypes, giving human experience a symbolic and inherited dimension. Their separation in 1913 marked a profound theoretical divergence, yet both thinkers arrived at a shared insight: much of what drives behaviour lies beneath conscious awareness, shaping our choices long before we can fully explain them to ourselves.

What makes these ideas matter is their reach beyond theory. The replicability crisis has reminded psychology that knowledge must be approached with humility, caution, and methodological rigour. In the courtroom, the stakes become unmistakably human: from M'Naghten Rules and occipital syndromes to landmark diminished responsibility cases such as Sutcliffe, Yates, Hinckley, Joyce & Kay, and Daryati, theories of mind can shape guilt, sentencing, and the meaning of responsibility itself. As neuroscience and neuroimaging continue to advance, they will further test the boundary between determinism and accountability, ensuring that the study of mind remains not only intellectually compelling, but ethically consequential.



What to Remember



System 1 Runs the Show

Most decisions are made automatically and unconsciously by System 1. Recognising when to engage System 2 is the foundation of better judgment.



Anchors Distort Every Estimate

The first number encountered skews all subsequent judgment — even when arbitrary. Professionals are not immune; awareness is the only defence.



Biases Are Systematic, Not Random

Confirmation bias, conjunction fallacy, sunk cost, framing effects, and availability heuristic are predictable errors — not exceptions. They follow patterns that can be learned and mitigated.



The Brain Is Modular

Discrete regions handle discrete functions — FFA for faces, V5 for motion, frontal lobes for executive control. Localised damage produces specific, predictable deficits.



The Unconscious Drives Behaviour

Freud showed that repressed drives shape thought and action below awareness. Libet's experiments confirmed the brain acts before conscious intention — by approximately 350ms.



Jung Expanded the Unconscious

Beyond the personal unconscious lies the collective unconscious — a universal layer of archetypes (Persona, Shadow, Anima/Animus, Self) shared across all humanity and cultures.



Freud and Jung Diverged Fundamentally

Freud saw the unconscious as pathological and sexuality-driven; Jung saw it as creative and meaning-centred. Their 1913 split gave birth to two enduring schools of depth psychology.



Free Will Is More Complex Than It Seems

Neural readiness potentials precede conscious decisions. Free will may operate as a veto or steering mechanism rather than an initiating force.



Science Must Earn Its Claims

The replicability crisis — where 11 of 12 priming studies failed replication — is a call for pre-registration, larger samples, and independent verification before publication.



Cognition Shapes Legal Outcomes

From M'Naghten Rules to Sutcliffe, Yates, Hinckley, and Daryati — theories of mind determine guilt, sentencing, and responsibility. Neuroimaging is now entering the courtroom.

Sources & Citations

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 **Note on Neuroscience Sources:** Brain specialisation findings (FFA, V4, V5, PPA) draw on the broader cognitive neuroscience literature including work by Nancy Kanwisher (MIT), Semir Zeki (UCL), and Roger Sperry's Nobel Prize-winning split-brain research (1981). The replicability crisis data references the R-Index analysis of Kahneman's Chapter 4 by Ulrich Schimmack (2012).